

KEYNOTE SESSION [TECH-401]

Advanced WebAssembly Runtimes & Edge Swarms

Architecting Low-Latency Autonomous Edge Intelligence with Sandboxed WASM



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#Wasm

#Rust

#EdgeSwarm

#DistributedAI

#LowLatency

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1. EDGE INGESTION

- Native WASM Bytecode compile
- Memory isolation via Wasmtime
- < 1.2ms cold start instantiation
- Zero JIT overhead on ARM/x86
- Direct hardware sensor binding

2. SWARM CONSENSUS

- Peer-to-peer WebRTC mesh
- CRDT state synchronization
- Ephemeral session memory
- Automatic network partition heal
- Cryptographic token exchange

3. AUTONOMOUS HEALING

- Closed-loop real-time monitor
- Hall capacity overflow alerts
- Dynamic schedule rescheduling
- Zero human intervention SLA
- Multi-agent arbitration matrix